

Mattuck Notes

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1 The Many-Body Problem (Ch. 0)

The many body problem can be thought of as the effect of interactions on system behavior, not belonging to any one subfield but rather an amalgamation of methods applicable to many-body systems in any field. Earlier methods included ignoring the interaction, or better yet choosing frames where interactions were negligible. But the advent of QFT changed things, introducing fictitious quasi particles that can be modeled in 'Feynman diagrams'.

Many body systems are complicated because they include many non-independent entities that interact with each other. Quasi particles, however, sidestep this issue as they are roughly non-interacting. Though this is not always the case, quasi particles can be thought of as a larger collection of particles, with a "main" particle and a cloud surrounding it consisting of other particles.

Think, for instance, an electron gas in a positively charged background (like a conducting metal). Here, the quasi particle consists of an electron, and the corresponding holes (bare positive background) that form around it due to Coulomb repulsion. The positive charges aren't moving with the electron, so as the electron moves the quasi particle's particular individual constituents change, but the quasi particle itself doesn't, as new holes join the quasi particle to replace the old ones.

Note that with many conducting electrons, they are all quasi-particles. It is reasonably clear they are roughly non-interacting unless the positively charged clouds overlap. For, at the boundary of the cloud, an electron must not feel any force from the quasi particle, since if it was attracted, the boundary would be closer, and if it was repelled, the boundary would be farther. This also means the positive charge surrounding the electron is equal to the charge of the electron, if we use the fact that a shell can be thought of as just a point at its center with the same charge. As we move further out from the boundary, there is no more accumulation of positive charge by definition, so the charge of the quasi particle is zero and hence there is no force from the quasi particle outside the boundary, and we can treat all of them as roughly independent.

However, as we've alluded already there's a problem if the clouds overlap, and in this case Coulomb repulsion is very significant and cannot be ignored. A quasi particle in the momentum state p will generally have some lifetime τ_p , the time before it interacts with another quasi particle (in this case, the clouds overlapping) and shifts to a new momentum state. Ideally we want this lifetime to be infinite since the point of using quasi particles is that they are non-interacting and non-interacting systems are easier to solve, but generally we just want it to be pretty long. Quasi particles also generally have a defined effective mass that takes into account the extra mass from the cloud of particles

surrounding the "bare" particle. This gives a new definition of the kinetic energy $p^2/2m$ with the new mass, whose difference with the main particle's kinetic energy is called the "self-energy".

There's also the idea of a "collective excitation", referring to the collective wavelike motion of all particles in a system. Some examples include plasmons (electron density waves in a metal foil when excited with high energy electrons), phonons (sound waves, i.e. lattice oscillation in solids) etc.

2 Feynman Diagrams (Ch. 1)

To actually find out what the lifetime and effective mass are, we need more. One way this can be done is by figuring out how a particle propagates through a system, which, from E&M we may recall as Green's functions, or more generally "propagators".

Propagators define the average behavior of a particle in a system, a function that takes an initial and final point in spacetime and calculates the probability that a particle (for a one-particle propagator) at the initial point ends up at the final point. One-particle propagators can find properties of quasi particles (and other stuff) and two-particle propagators can find properties of collective excitations (and other stuff). Vacuum can be examined using the no-particle propagator.

As a simple example Mattuck introduces the drunken man propagator, where the man (M) plays the part of a particle and the bars as a sort of background for the particle to interact with. He starts at point 1 (P1) and the end goal is point 2 (P2), but along the way he can go to various bars but let's deal with two, call them A and B (Mattuck has a third bar C he never actually uses so I'll omit it).

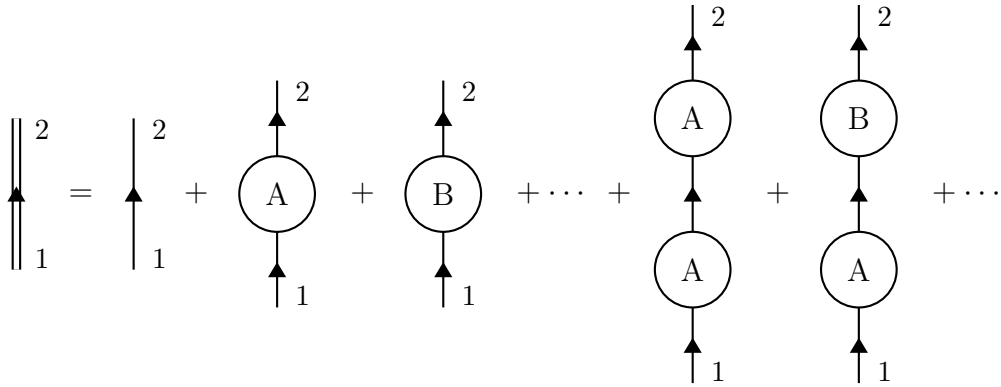
Now, going from site 1 to site 2 directly is called "free propagation", denoted $P_0(2, 1)$, whereas including all possible routes with interactions included is encoded by the propagator, denoted $P(2, 1)$. Each site (P1, P2, A, B), have free propagators between them, i.e. probabilities for M to go between them, and M has a probability of drinking at (interacting with) the bars as well. All of these probabilities are taken to be independent for simplicity, so that the probability of, say, M going from P1 to B, drinking at B and then going from B to P2 is just $P_0(B, P1)P(B)P_0(P2, B)$.

The full propagator includes the free propagator, and then adds on nontrivial paths where interactions are included. The probabilities of each path simply sum as we are used to to yield a total probability assigned to the propagator. The expansion will look like so:

$$P(2, 1) = P_0(2, 1) + P_0(A, 1)P(A)P_0(2, A) + P_0(B, 1)P(B)P_0(2, B) + \dots \quad (1)$$

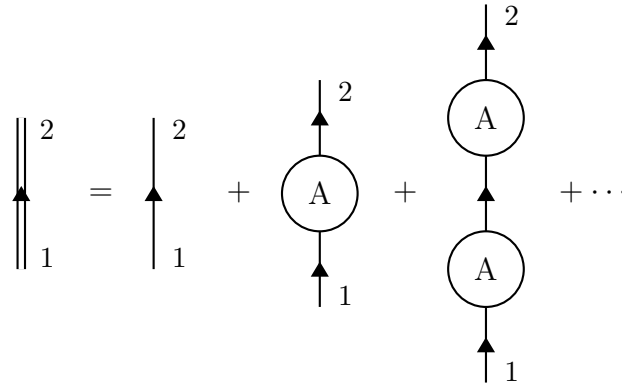
where the ellipses will include second order terms where, to use the analogy, the drunken man visits A, drinks, visits A again, drinks, and then goes home, or goes from drinking at A to drinking at B and then goes home, etc. In the same way, all higher order terms will also be included.

Pictorially we can write it like so:

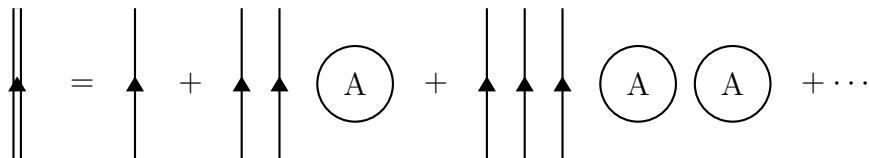


where the interpretation is to think of the double line as the total propagator, a single line as a free propagator, and the nodes as the probability of interaction. The first ellipses is kind of redundant because every first order term is encapsulated there for this system, but more generally when there are more bars/sites that will include other terms as well. The second ellipses includes the other omitted second order terms as well as higher order terms.

One popular way to simplify such a diagram is to think of only bar A is a significant term, wholly omitting anything involving B. This yields a diagram like so:



To further simplify things, for instance, we can assume each free propagator has the same probability, at least in relation to A (so the probability the drunken man goes from 1 to 2, 1 to A, A to 2, A to A are all the same). And recall that since all the probabilities are independent we can treat it as a simple multiplication of the individual parts of the diagram. This means we can write this diagram like so:



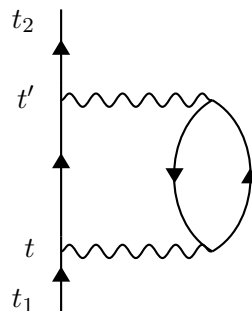
Now, the propagators are probabilities which are smaller than or equal to one (but smaller for a nontrivial system), and so we see this is actually a geometric series in the free propagator that actually converges. In fact we can take one of the free propagators in common among them all, and then use the classic infinite geometric series result to write:

$$\parallel \uparrow = \uparrow \times \left(\frac{1}{1 - \uparrow \text{---} \textcircled{A}} \right) = \frac{1}{\uparrow \text{---}^{-1} \text{---} \textcircled{A}}$$

The drunken man propagator and all its other forms and approximations apply exactly to real physical systems like electrons in a metal. In particular, the free propagator is just the quantum mechanical probability amplitude for a particle to go from some point in the metal to another without interaction (also at specified times), and the probability of drinking at a bar is translated to the probability amplitude of interacting with a positive ion in the metal. With this dictionary, every Feynman diagram above applies directly to the electron in a metal system.

We can now move onto many body systems, and consider a single-particle propagator. It should be noted that these are quasi particles, and the interpretations are strictly for convenience and not physical. For the time being we can also think of the system as an electron gas in a positive background like in a metal, and we can think of a volume/area with the electron "lifted out" as a "hole". Both particles and holes are drawn in Feynman diagrams.

The single-particle propagator can roughly be interpreted as the propagator for an extra electron being shot into the metal. Obviously this will cause some disturbance, displacing electrons and forming holes, and there are many different ways it can do this. One simple interaction is by having the electron lift out one of the electrons already in the system, propagate freely with both the hole and lifted electron to some other point, and then push the lifted electron back into the hole before exiting the system. This is a "second order process" because it has two interactions, and it can diagrammatically be presented like so (I'll explain the meaning of the symbols after):

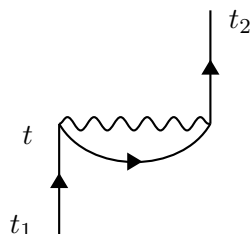


In Feynman diagrams, time moves forward as you go up, and the single lines have the same interpretation as before (free propagation). The oval bubble there is a "particle-hole" bubble. A hole is represented just like a particle but in the opposite direction for... reasons I'm not privy to yet. The squiggly line shows that there's an interaction taking place between the two particles at that time.

So, what this diagram is saying is that there is some particle entering the system at t_1 , interacts with another particle in the system at t , forming a particle-hole pair, and everything free propagates until they interact again at t' , out of which only the original

particle emerges out at t_2 , implying the particle-hole pair vanished. So essentially it says the same thing as the situation we described above.

There are also first order processes where a particle entering just takes the spot of an existing electron and that electron gets shot out of the system, and that has a diagram like so:



The interpretation here is that some extra particle emerges at t_1 , interacts with some particle in the system at t , after which the extra particle goes to the location of the system particle, which emerges out of the system at t_2 . This shape is called the "open oyster". I suspect there's more to the story since it looks like the lower lip is a particle going against time.

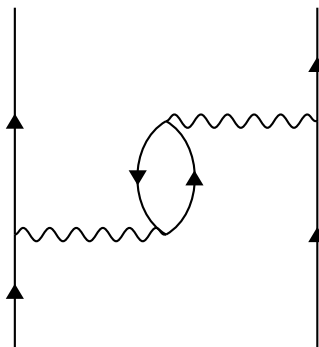
The open oyster is just one diagram in a series of many diagrams that are added up to form the one particle propagator (this series also includes the second order term we talked about earlier). I won't bother writing it all out here since they aren't really explained in the text, more so just presented. But the point is that all of these individual diagrams are possibilities of the form of the quasi particle, and how it may evolve, which is sort of what we were looking for.

Like before we can actually perform the same trick on the single-particle propagator, only including terms that are a product of open oysters like so:

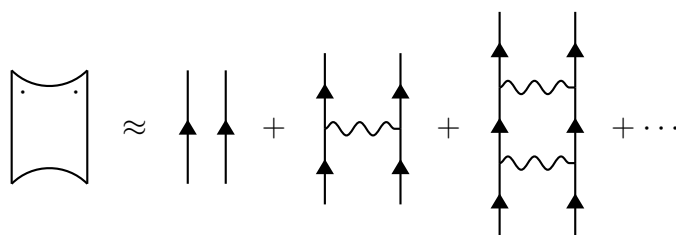
$$\begin{aligned}
 \text{Diagram 1} &= \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} + \dots \\
 &= \left[\text{Diagram 2} - \text{Diagram 5} \right]^{-1}
 \end{aligned}$$

where in analogy all we've done is only consider bar A, and then use the independence of events to treat each diagram as the product of its constituents, take stuff in common, and use the geometric series result. What we've written above is actually the Hartree-Fock approximation. There are other approximations he shows but he doesn't really talk about them so I'll omit it here.

There is also the two-particle propagator. This propagator will involve things like each particle doing the things we've already discussed for the single particle propagator, but it will also include terms where the two particles of interest are interacting either directly or "virtually" i.e. via a particle in the system. Such a "virtual" term would look like so:



For direct interactions there just wouldn't be a bubble in the middle, instead the diagram would look like a "ladder". In fact this motivates the name "ladder approximation", which essentially neglects terms like the one I've shown above. The series ends up looking like so:



There are some other approximations but I won't list them here since he doesn't really talk about them at any detail anyway.

Finally there's the no particle propagator that can deal with vacuum and also find the ground state energy of a system. There can be situations like particles switching places (a first order process) where the system doesn't just trivially sit there, and so this system has its own series of Feynman diagrams. Again I don't really think there's any crucial information here other than just stuff to look at so I won't bother trying to recreate these Feynman diagrams... I'll deal with it when it becomes necessary.

3 Quasi Particles & The Pinball Propagator (Ch. 2)

This chapter is largely a repeat of the previous chapter but for a new system, a pinball machine, where a pinball enters at spacetime coordinate $(\mathbf{r}_1, t_1)$ and has the chance to interact with various objects in the machine before ending up at some final location $(\mathbf{r}_2, t_2)$. Technically it could end up elsewhere at a different time, but the idea is to try to figure out the probability (amplitude) of the pinball ending up at this final spacetime coordinate.

Previously we've dealt with the time-independent case and the same discussion follows here, so I won't repeat it. Mattuck does cover a new type of problem with the same "strong interaction" regime we've discussed, but without assuming all the free propagators have the same probability amplitude, but again this is a pretty trivial discussion and the way you solve it is identical, with a couple tweaks. This "strong interaction" regime (where you only include diagrams with the strong interactions and none of the others) is one example of "partial/selective summation", a method more powerful than perturbation theory where all you do is work to some finite order, often first or second, and ignore the rest, which obviously wouldn't do great if one interaction was particularly strong.

The time-dependent case is no different diagrammatically, and all we do is label each point both with a spatial and a time coordinate. The interaction terms (circles with a letter inside) will also pick up a time coordinate because they happen at some specific time now. In general, the interaction time can be any time between the entry and emergence of the pinball, and we must account for the probability amplitudes of the pinball interacting with said object at every possible time in between t_1 and t_2 .

There are a couple details that are necessary to discuss before any calculations. First, as before, any interaction between the pinball and some object A is $P(A)$, and happens at time t_A at location $\mathbf{r}_A$ (we'll put primes on the spacetime location if there are multiple interactions with A). When such an interaction occurs, the particle loses memory of how it got there or any of its past. This physically enforces our previously held assumption that the various parts of the diagram are independent in the probabilistic sense.

The probability of a particle starting at $(\mathbf{r}_1, t_1)$ and ending up at $(\mathbf{r}_2, t_2)$ is labeled $P(\mathbf{r}_1, \mathbf{r}_2, t_2 - t_1)$, where I've coalesced the time coordinate as Mattuck does later anyway. This is another assumption we're making: time translation invariance (the force that makes the particle wade through the system, (in a regular pinball machine this is gravity) is constant, so only differences in time matter). If you slap on a subscript of 0, it becomes the free propagator as before.

Now, the propagator can be written like so (take G, M as objects in the system, among others):

$$\begin{aligned}
P(\mathbf{r}_1, \mathbf{r}_2, t_2 - t_1) &= P_0(\mathbf{r}_1, \mathbf{r}_2, t_2 - t_1) \\
&+ \int_{t_1}^{t_2} dt_G P_0(\mathbf{r}_G, \mathbf{r}_1, t_G - t_1) P(G) P_0(\mathbf{r}_2, \mathbf{r}_G, t_2 - t_G) \\
&+ \int_{t_1}^{t_2} dt_M P_0(\mathbf{r}_M, \mathbf{r}_1, t_M - t_1) P(M) P_0(\mathbf{r}_2, \mathbf{r}_M, t_2 - t_M) \\
&+ \dots \\
&+ \int \int \dots \\
&+ \dots
\end{aligned}$$

The first set of dots includes other first order terms for the other objects just like for G and M, while the second set of dots includes second order terms that include interacting with two objects (including the same one twice), and that would be a path like going to

G, traveling back to G, and then exiting, or going to G, traveling to M, and then exiting, etc. The last set of dots includes all higher order terms.

Again, nothing is new about this and we've seen it already, just in a time-independent form. Now there are integrals we have to slap on because t_G can happen whenever between t_1 and t_2 , and so we have to add up all the ways in which G can be interacted with to find the total propagator. Same story with the other objects.

Now this integral business is a mess, but we can fortunately cast it in a nicer way if we use the Fourier transform, i.e. change it to a "frequency domain" as the engineers call it. To perform a Fourier transform our integrals must be from $-\infty$ to ∞ , which we can facilitate if we define all values of the probability where the time coordinate is negative to be zero. This way if t_G , say, is over t_2 , then $t_2 - t_G$ will be negative and the integral vanishes, and if t_G is below t_1 then $t_G - t_1$ will be negative and the integral still vanishes. Same story for the rest.

In taking the Fourier transform, time is integrated out and its role is replaced with a frequency. It's also linear so we can treat each term as its own in the above perturbative calculation. I'll use the Fourier transforms of each function with the same label, but with just the time coordinate replaced with a frequency (without the "hat" that's usually used). With this convention nothing special happens with the free propagator, but some nontrivial calculations can be done with the first order term that you can extrapolate to the higher order terms.

I'll go through the gist of it:

$$\begin{aligned}
& \int_{t_1}^{t_2} dt_G P_0(\mathbf{r}_G, \mathbf{r}_1, t_G - t_1) P(G) P_0(\mathbf{r}_2, \mathbf{r}_G, t_2 - t_G) \\
&= \int_{-\infty}^{\infty} dt_G P_0(\mathbf{r}_G, \mathbf{r}_1, t_G - t_1) P(G) P_0(\mathbf{r}_2, \mathbf{r}_G, t_2 - t_G) \\
&= \int_{-\infty}^{\infty} dt_G \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' e^{-i\omega'(t_G - t_1)} P_0(\mathbf{r}_G, \mathbf{r}_1, \omega') \right] \\
&\quad \times P(G) \times \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega(t_2 - t_G)} P_0(\mathbf{r}_2, \mathbf{r}_G, \omega) \right]
\end{aligned}$$

In the first step, as discussed, I extended the bounds to cover the entire real line, and in the second I've replaced the time dependent functions with how you'd get them from their Fourier transformed variants (i.e., by just replacing them with the inverse Fourier transform applied on their Fourier transform). Here we note that only the exponential terms have a t_G dependence, and so isolating that and evaluating the integral looks like $\int_{-\infty}^{\infty} dt_G e^{-it_G(\omega' - \omega)} = 2\pi\delta(\omega' - \omega)$.

To actually derive this identity is... tricky. You can do it by using regularization, which is to basically take that integral (which obviously doesn't converge) and add on a term in the exponent, like $-\lambda t^2$, that makes the integral converge, and then take $\lambda \rightarrow 0$. Now, obviously there's some mathematical voodoo going on and this is not, even slightly, rigorous, but apparently if we treat all of these objects not just as functions but as probability distributions, this kind of trickery is justified. At least, this is the conclusion I came to back when I first saw it and reading up online.

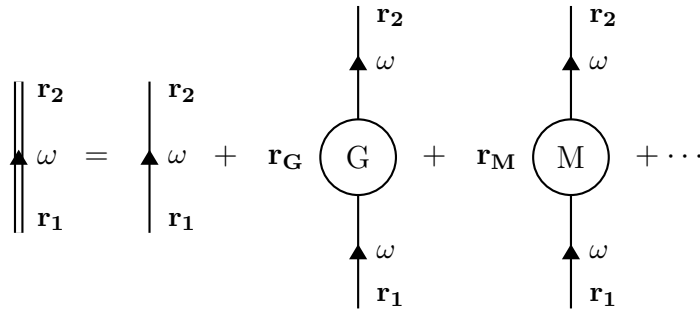
Anyway once you have that identity you can easily plug it into that integral expression above and derive:

$$\begin{aligned} \int_{t_1}^{t_2} dt_G P_0(\mathbf{r}_G, \mathbf{r}_1, t_G - t_1) P(G) P_0(\mathbf{r}_2, \mathbf{r}_G, t_2 - t_G) \\ = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega(t_2-t_1)} P_0(\mathbf{r}_G, \mathbf{r}_1, \omega) P(G) P_0(\mathbf{r}_2, \mathbf{r}_G, \omega) \end{aligned}$$

which is just the inverse Fourier transform of that integrand without the exponential bit. So, if we were to take a Fourier transform of our integral mess from earlier, we can write:

$$P(\mathbf{r}_2, \mathbf{r}_1, \omega) = P_0(\mathbf{r}_2, \mathbf{r}_1, \omega) + P_0(\mathbf{r}_G, \mathbf{r}_1, \omega) P(G) P_0(\mathbf{r}_2, \mathbf{r}_G, \omega) + \dots$$

which you will recall is precisely the same expansion as the time-independent case in the time domain (see (1)). In this domain the Feynman diagrams are slightly tweaked, and are written as follows:



where I've omitted terms of second order or higher since we've already seen it. Basically all we're doing is labeling the interactions and putting an ω next to the propagators.

I will now move onto solving the first three problems given in the textbook. They're really just three parts of the same problem so I'll treat them as such.

Problem

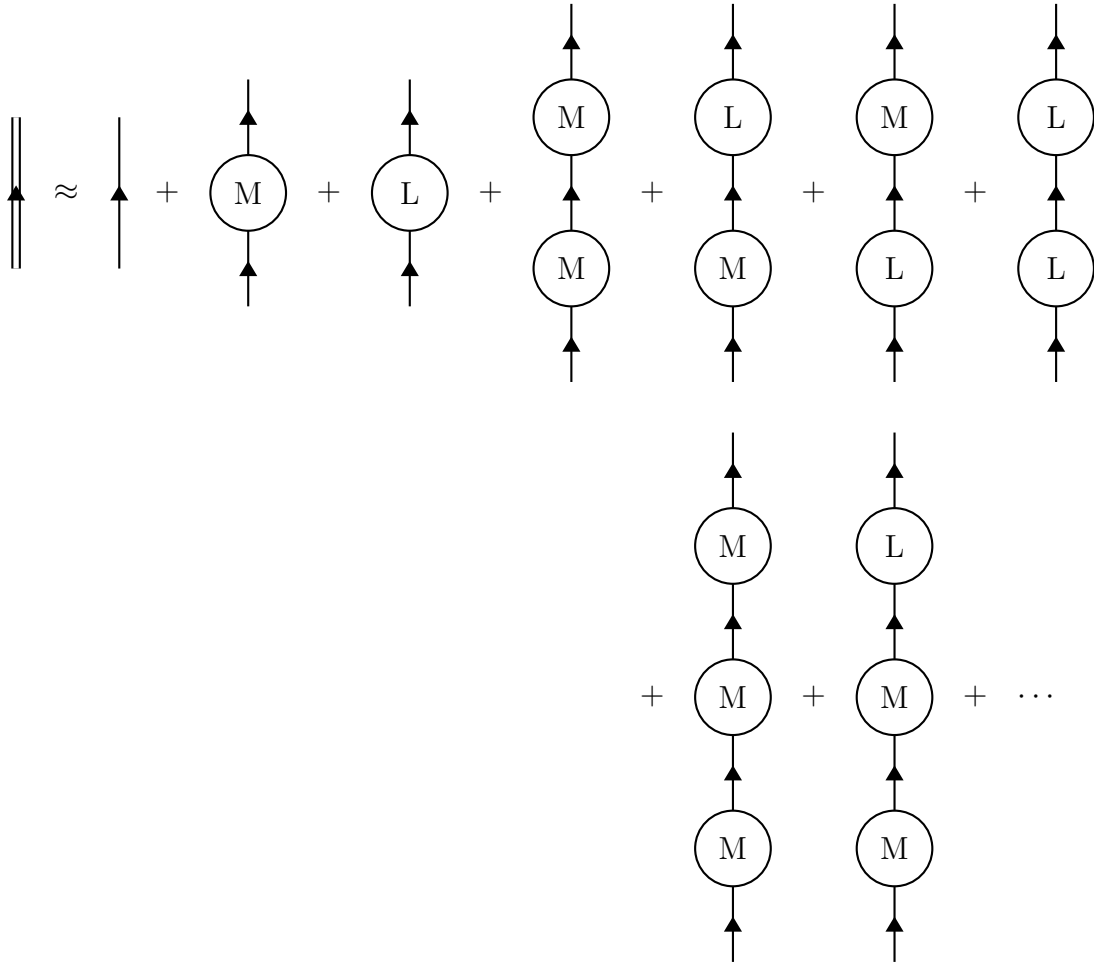
a) Write the diagram series for the propagator $P(\mathbf{r}_2, \mathbf{r}_1)$ where the scattering due to objects M and L are large whereas other objects interact very weakly. Include terms up to second order and a couple of third order terms. How many diagrams are there in the n -th order?

b) Write this series in functional form.

c) Evaluate the propagator via partial summation assuming all free propagators have the same probability (amplitude) c .

Solution

a) The idea is to use the strong interaction approximation but to include all terms where only the two strong interactions are there. So this will be all the terms that would've been there if only M was strong, all the terms that would've been there if only L was strong, and all the terms that mix M and L. To second order plus a couple extra terms we find:



The pattern is clear enough, if you want to think of it systematically I guess you can think of it as counting in binary and treating "M" and "L" as "1" and "0". If there's more objects it would just be counting in a different base and assigning each object its own number. Also, with some basic combinatorics we surmise there are 2^n different diagrams at n -th order. If you had k strong interactions there would be k^n diagrams at n -th order.

b) I'll just do the first order and a couple of the second order terms.

$$\begin{aligned}
P(\mathbf{r}_2, \mathbf{r}_1) &\approx P_0(\mathbf{r}_2, \mathbf{r}_1) + P_0(\mathbf{r}_M, \mathbf{r}_1)P(M)P_0(\mathbf{r}_2, \mathbf{r}_M) + P_0(\mathbf{r}_L, \mathbf{r}_1)P(L)P_0(\mathbf{r}_2, \mathbf{r}_L) \\
&+ P_0(\mathbf{r}_M, \mathbf{r}_1)P(M)P_0(\mathbf{r}_M, \mathbf{r}_M)P(M)P_0(\mathbf{r}_2, \mathbf{r}_M) \\
&+ P_0(\mathbf{r}_M, \mathbf{r}_1)P(M)P_0(\mathbf{r}_L, \mathbf{r}_M)P(L)P_0(\mathbf{r}_2, \mathbf{r}_L) \\
&+ \dots
\end{aligned}$$

c) Now every P_0 term evaluates to c . So we can just think of each free propagator line as some constant we can pull out, and only consider the bubbles within each order. At n -th order all we're doing is exhaustively pairing up (i.e. multiplying) n interactions where each interaction can be either M or L, and then adding them up. But, well, that is exactly what the middle schoolers call "foiling", i.e. it's just $[P(M) + P(L)]^n$. With that in mind it's pretty easy to see:

$$\begin{aligned}
P(\mathbf{r}_2, \mathbf{r}_1) &\approx c \sum_{n=0}^{\infty} c^n [P(M) + P(L)]^n \\
&= c \cdot \frac{1}{1 - c[P(M) + P(L)]} \\
&= \frac{1}{c^{-1} - [P(M) + P(L)]}
\end{aligned}$$

which is the same result as the strong interaction for one object, but with the probability of the interaction replaced with the sum of the two strong interaction probabilities.

4 The Quantum Analogue (Ch. 3)

As the name suggests this is essentially repeating all we've been doing but in the lens of quantum mechanics, and generally treating the topics in a more explicit way. Strictly speaking the system we consider isn't any different, but the way we solve it does notably change, though as we'll see in the diagram formalism it is actually something we've already seen before.

The first significant departure from what we have thus covered is that we no longer work with plain probabilities, but rather probability amplitudes, which are quantities that yield probabilities when you take a modulus square. In quantum mechanics individual processes affect each other and so the probabilities do not simply add. However, the amplitudes do add, and the probability for a process is received just by taking the modulus squared of the sum of all the amplitudes.

The amplitudes are denoted by $iG(\mathbf{r}_2, \mathbf{r}_1, t_2 - t_1)$ where the factor of i is simply convention. This function represents the probability amplitude of a particle added to the (interacting) system at $(\mathbf{r}_1, \mathbf{t}_1)$ ending up at $(\mathbf{r}_2, \mathbf{t}_2)$, while the system itself is in its ground state at t_1 and t_2 . We'll talk briefly later on how these probability amplitudes are actually

Green's functions, but if we were to briefly accept that, we should note that the function is going to be defined for all time since it's more convenient, and for the "retarded" propagator, negative time components yield zero, while for the "advanced" propagator the negative time components are nontrivial. We'll only work with the retarded propagator for this section.

Ultimately we want to use the language of QM to describe things a bit more concretely, and a good place to start is sensibly using the free particle Hamiltonian solutions to describe the free propagator. In between every interaction we will have a free particle, and so a notation that takes advantage of being able to label these (momentum eigen-) states is beneficial.

This leads to the notation $iG^+(\mathbf{k}_2, \mathbf{k}_1, t_2 - t_1)$ which has the same meaning, except that we think of adding the extra particle in the free particle eigenstate $\phi_{k_1}(\mathbf{r})$ instead of the position eigenstate represented by $\mathbf{r}_1$, and find the amplitude of it ending up in the final free particle eigenstate $\phi_{k_2}(\mathbf{r})$. I don't think of it as much more than a basis change. Also note the $+$ on the propagator, which stands for "retarded". This will make it so that all negative time components yield zero as discussed previously.

The goal now is to figure out an expression for the free propagator now that we have a quantitative way to describe it (i.e. it has a diff. eq. we can solve). If we were to start in the state $\phi_{k_1}(\mathbf{r})$ at time t_1 and let the system freely propagate, we know from the time-dependent Schrödinger equation that the wavefunction simply picks up a complex exponential factor, and at time t_2 becomes $\phi_{k_1}(\mathbf{r})e^{-i\epsilon_{k_1}(t_2-t_1)}$, where ϵ_k is the free particle energy corresponding to quantum number k (or, if you prefer, the momentum in units of $\hbar$). For the free particle the wavefunctions and energies are given as follows ($\hbar = 1$):

$$\phi_k(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}} \quad , \quad \epsilon_k = \frac{k^2}{2m}$$

where Ω is a normalization constant representing volume. To find the propagator we need to figure out the component of the wavefunction at t_2 along some final free particle eigenstate (with wavenumber k_2), which is trivially just $e^{-i\epsilon_{k_1}(t_2-t_1)}\delta_{k_2k_1}$ if we use the orthonormality of the eigenfunctions. Now if we simply tack on a factor that ensures this factor is zero for $t_2 < t_1$, we have our free propagator, and this service is fulfilled by the Heaviside step function $\theta_{t_2-t_1}$.

All in all we surmise:

$$\begin{aligned} G_0^+(k_2, k_1, t_2 - t_1) &= -i\theta_{t_2-t_1} e^{-i\epsilon_{k_1}(t_2-t_1)} \delta_{k_2k_1} \\ &= \delta_{k_2k_1} G_0^+(k_1, t_2 - t_1) \end{aligned}$$

where we've defined

$$G_0^+(k, t_2 - t_1) = \begin{cases} -i\theta_{t_2-t_1} e^{-i\epsilon_k(t_2-t_1)}, & t_2 \neq t_1 \\ 0, & \text{otherwise} \end{cases}$$

From here on forth, in this section at least, we will treat this quantity as the free propagator. The idea here is that the two wavenumbers k_2, k_1 are redundant. The reason

for this is that the systems we will talk about in this section are translationally invariant, and momentum, as the generator of translation, is conserved (by Noether's theorem classically). So ultimately what we'll end up seeing is that the total propagator is just this "reduced" propagator multiplied by a delta function that forces the final wavenumber to be the same as the initial. At the end of this section we'll consider a system where this isn't completely true, but we'll cross that bridge when it comes to it.

We've already seen that transforming a structure like this to Fourier space is worthwhile, so let's do that. We wish to evaluate the expression:

$$G_0^+(k, \omega) = -i \int_{-\infty}^{\infty} d(t_2 - t_1) \theta_{t_2 - t_1} e^{i\omega(t_2 - t_1)} e^{-i\epsilon_k(t_2 - t_1)}$$

It doesn't take a genius to figure out this doesn't converge, so we have to use some trickery, particularly "regularization". The problem right now is that the function isn't dying off fast enough at large times, so we modify the expression like so:

$$G_0^+(k, \omega) = -i \int_{-\infty}^{\infty} d(t_2 - t_1) \theta_{t_2 - t_1} e^{i\omega(t_2 - t_1)} e^{-i\epsilon_k(t_2 - t_1)} e^{-\delta(t_2 - t_1)}$$

where δ is an infinitesimally small positive number. So long as it is not zero, it will make this integral converge, and it's trivial to show that:

$$G_0^+(k, \omega) = \frac{1}{\omega - \epsilon_k + i\delta} \quad (2)$$

Note that the pole of this function corresponds to the energy of the free particle. This is actually a general result: the poles of a propagator (general, not just free) occur at values of ω equal to the excited state energies of the (N+1 body) system minus the ground state energy of the (N body) system. We will in fact use this for a problem at the end of this section.

Before we embark on more explicit calculations it is worth revisiting quasi particles and what kind of structure we expect its propagator to have. Well, we know it has a "dressed" energy/mass ϵ'_k , and a lifetime τ_k . We can roughly guess that this lifetime appears exponentially, i.e. a particle with wavenumber k will fade out of this momentum state via exponential decay. Accounting for fermionic corrections for the particle clouds and such we also add a correction factor Z_k , yielding the space-time form:

$$G_{\text{quasi}}^+(k, t_2 - t_1) = -i Z_k e^{-i\epsilon'_k(t_2 - t_1)} e^{-(t_2 - t_1)/\tau_k}$$

which has Fourier transform:

$$\frac{Z_k}{\omega - \epsilon'_k + i\tau_k^{-1}} \quad (3)$$

This is already similar to (2), and we can thus surmise we probably want the inverse of the lifetime to be like δ , which is infinitesimal, which is to say the lifetime should be

very long as we expect. There's a little discussion on interpreting this lifetime term as part of a complex energy but I won't adapt it since it looks like a tangent.

Now we go on to treat an actual system, the quantum variant of the pinball system. This time the objects are not rigid objects to interact with necessarily, but rather potentials that can affect the particle. For simplicity we'll only consider two potentials at the moment, one proportional to p^2 , another to p^4 , yielding the total Hamiltonian:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \underbrace{M\hat{p}^2}_{\hat{V}_M} + \underbrace{L\hat{p}^4}_{\hat{V}_L} \quad (4)$$

where for illustrative purposes we also take $M \gg L$. A system like this can be found with COM motion, relativistic corrections, magnetic fields, etc.

Anyway, what we'd like to do is to find the energies of this system subject to these perturbations, first in the way we'd usually do it and then using a propagator formalism.

The conventional solution is instant: the eigenfunctions are clearly just the eigenfunctions we've already been working with (i.e. momentum eigenstates), and each instance of $\hat{p}$ just adds a factor of k . Defining ϵ'_k as the energy if we were to ignore L , and ϵ''_k as the total energy, we find:

$$\epsilon'_k = \left(\frac{1}{2m} + M \right) k^2 \quad , \quad \epsilon''_k = \left(\frac{1}{2m} + M \right) k^2 + Lk^4 \quad (5)$$

Let's now use propagators. Again, the idea is that the propagator adds up all the various paths the particle can take through the system: completely free propagation, being scattered once (by M or L), twice (MM, ML, LM, LL), etc. One thing we didn't really bother with the details on was the "interaction probability", which we can hazard a good guess for here using time dependent perturbation theory. To put things briefly, the probability amplitude per unit time that the system goes from wavenumber k_1 to k_2 (due to an instantaneous interaction) is equal to $-iV_{k_1 k_2}$ where the indices correspond to taking the inner product with those wavefunctions. This is easy to actually calculate but it's not pertinent to our discussion so I'll leave it out. The important point is that the potentials conserve momentum, which can be seen by noting the delta function that shows up in the result.

So the term that captures the probability amplitude of going from the initial state, to being scattered by V_M , and then going to state k_2 is given by:

$$G_{t_1 \rightarrow t_M \rightarrow t_2} = i \int_{-\infty}^{\infty} dt_M G_0^+(\mathbf{k}_2, t_2 - t_M) V_{M k_2 k_1} G_0^+(\mathbf{k}_1, t_M - t_1)$$

Same kind of idea with V_L . Recall that we integrate over t_M since it can occur any time between t_1 and t_2 . The final perturbative series for the total propagator is akin to the integral mess from before with the probabilities, where the integrals essentially encode paths through the system. More importantly, in taking the Fourier transform the propagator ends up taking the following form (omitting $(\mathbf{k}, \omega)$):

$$\begin{aligned} G^+ = G_0^+ &+ [G_0^+]^2 V_{Mkk} + [G_0^+]^2 V_{Lkk} + [G_0^+]^3 V_{Mkk}^2 + 2 [G_0^+]^3 V_{Mkk} V_{Lkk} \\ &+ [G_0^+]^3 V_{Lkk}^2 + \dots \end{aligned}$$

which you'll note is something we've already seen as part of a problem from the last section. At this point you can, as we have already done before, go to a diagram formalism and treat each G_0^+ factor as a free propagator, each V term as a bubble, and then use the same trickery. This is not new so I won't include it. However, one new interesting observation is that the perturbative diagram series can be written like so (ignoring L for the moment, since it is much smaller than M anyway):

$$\text{Double line with } \uparrow = \text{Single line with } \uparrow + \text{Single line with } \uparrow \text{ through bubble } M$$

You can check this recursively: plugging it back in, we see in each recursive step you get one term replacing the total propagator with the free propagator under the bubble, and another term that adds another bubble with the total propagator called recursively again. Going on and on, we get the strong interaction approximation for one strong interaction as we've seen before. What's nice about this is that we can subtract the second diagram on the right on both sides, take the total propagator as common, and then rearrange to find:

$$\begin{aligned} \text{Double line with } \uparrow \times \left\{ 1 - \text{bubble } M \right\} &= \text{Single line with } \uparrow \\ \Rightarrow \text{Double line with } \uparrow &= \left[\text{Single line with } \uparrow^{-1} - \text{bubble } M \right]^{-1} \end{aligned}$$

which is of course already a result we have seen. In the present context we can write it in a functional form:

$$G^+(\mathbf{k}, \omega) = \frac{1}{(G_0^+)^{-1} - V_{Mkk}} = \frac{1}{\omega - (\epsilon_k + Mk^2) + i\delta} \quad (6)$$

where I've used that $V_{Mkk} = Mk^2$ which is fairly clear. Comparing this expression to the quasi particle expression (3), we see that:

$$\epsilon'_k = \epsilon_k + Mk^2 = \left(\frac{1}{2m} + M \right) k^2 \quad (7)$$

$$\tau_k = \frac{1}{\delta} = \infty \quad (8)$$

The energy is just what we expected and the lifetime is what we'd like to see for a quasi particle, and evidently true since momentum is conserved and the particle will never get knocked into a different momentum eigenstate.

Including L is nothing special, it's just like the problem we solved earlier relating to two strong interactions. The result is just:

$$\text{Diagram: double vertical line with upward arrow} = \left[\text{Diagram: single vertical line with upward arrow}^{-1} - \left(\text{Diagram: circle with M} + \text{Diagram: circle with L} \right) \right]^{-1}$$

Again, same story on the functional side of things:

$$G^+(\mathbf{k}, \omega) = \frac{1}{(G_0^+)^{-1} - (V_{Mkk} + V_{Lkk})} = \frac{1}{\omega - (\epsilon_k + Mk^2 + Lk^4) + i\delta} \quad (9)$$

and again for the resulting quasi particle we see:

$$\epsilon_k'' = \epsilon_k + Mk^2 = \left(\frac{1}{2m} + M \right) k^2 + Lk^4 \quad (10)$$

$$\tau_k = \frac{1}{\delta} = \infty \quad (11)$$

This is all in agreement with what we knew to obviously be true from basic QM. But this method is certainly overkill for such a simple system, so we will in a moment consider a trickier and nontrivial system.

In general we have largely been using classical analogies to construct the formalism with some basic QM computations, but the perturbative series of diagrams can be entirely derived in the lens of QM alone. This isn't too important for what's about to come so I'll just briefly describe it without going into details. The idea is to first take the diff. eq. that the Green's function should solve (i.e., Schrödinger operator, delta function source), apply a Fourier transform to find the diff. eq. in k-space. At this point you can show that a recursive integral relation solves this diff eq., which can be successively expanded using the recursion relation to generate the perturbative series.

Let's move onto an actual system now: an electron in an impure metal. Forget the background, and only consider the impure ions (of which there are N), assumed to be identical and randomly distributed in the metal's volume, Ω . The Feynman diagram series looks exactly as before: free propagators going into various interaction combinations. We assume the i -th impurity has a location $\mathbf{R}_i$, with potential $W(\mathbf{r} - \mathbf{R}_i)$.

This time the interaction term is (where k is the initial wavenumber and l is the wavenumber after the interaction):

$$V_{lk}(\mathbf{R}_i) = \frac{1}{\Omega} \int d^3\mathbf{r} e^{-i(l-\mathbf{k})\cdot\mathbf{r}} W(\mathbf{r} - \mathbf{R}_i) = \frac{1}{\Omega} e^{-i(l-\mathbf{k})\cdot\mathbf{R}_i} W_{lk} \quad (12)$$

where we've used:

$$W_{lk} = \int d^3\mathbf{r}' e^{-i(\mathbf{l}-\mathbf{k})\cdot\mathbf{r}'} W(\mathbf{r}') \quad (13)$$

after changing variables to $\mathbf{r} - \mathbf{R}_i$. We write the functional form for the propagator like so (omitting ω):

$$\begin{aligned} G^+(\mathbf{k}_2, \mathbf{k}_1) &= G_0^+(\mathbf{k}_1) \delta_{k_1 k_2} + G_0^+(\mathbf{k}_2) \sum_{i=1}^N V_{k_2 k_1}(\mathbf{R}_i) G_0^+(\mathbf{k}_1) \\ &+ G_0^+(\mathbf{k}_2) \left[\sum_l \sum_{i=1}^N V_{k_2 l}(\mathbf{R}_i) G_0^+(\mathbf{l}) V_{l k_1}(\mathbf{R}_i) \right] G_0^+(\mathbf{k}_1) \\ &+ G_0^+(\mathbf{k}_2) \left[\sum_l \sum_{j \neq i} V_{k_2 l}(\mathbf{R}_j) G_0^+(\mathbf{l}) \sum_{i=1}^N V_{l k_1}(\mathbf{R}_i) \right] G_0^+(\mathbf{k}_1) + \dots \quad (14) \end{aligned}$$

The interpretation for the first couple terms is rather clear: the first term is just free propagation, the second term considers scattering at one impurity for each impurity present.

The third term is somewhat ugly but it represents scattering at the same impurity twice, for every impurity. To see this, going from the right we see free propagation in state $\mathbf{k}_1$, interaction with the potential that bumps the particle to state $\mathbf{l}$, free propagation in this state, then finally another interaction at the same spatial point that bumps the particle into $\mathbf{k}_2$. Now, this can happen at every impurity location so we sum over i , but at the same time in between the interactions, the particle could have taken on any intermediate state, so we must sum over l as well.

The fourth term represents scattering at two different impurity ions, and the breakdown goes similarly. There is a plain sum over every impurity ion, and then a sum over every impurity ion which is *not* that ion, and again the same sum over $\mathbf{l}$.

Now, strictly speaking this is not translationally invariant. And it is cumbersome to try to find an exact solution for a particular choice of $\{\mathbf{R}_i\}$. What we can do to ease our troubles is to try to find the *average* of the propagator. We can imagine that if the impurity ions are randomly distributed, and we average over every possible arrangement of the impurity ions, it almost mimics a uniformly distributed system of impurity ions that does have translational invariance, which then means we might hope to still see momentum conservation. Another good reason to do this is that the variation from the mean propagator vanishes (relative to the size of the mean), which is to say, usually, the actual propagator is pretty close to the average propagator anyway for a large enough system. Let's find the averages of the first few terms we've presented in (14) to get an idea of what we're dealing with.

The first term is clear, since there is no $\mathbf{R}_i$ dependence. The second term goes like follows:

$$\begin{aligned}
\left\langle G_0^+(\mathbf{k}_2) \sum_{i=1}^N V_{k_2 k_1}(\mathbf{R}_i) G_0^+(\mathbf{k}_1) \right\rangle &= G_0^+(\mathbf{k}_2) G_0^+(\mathbf{k}_1) \cdot N \left\langle \langle V_{k_2 k_1}(\mathbf{R}_i) \rangle_{\{\mathbf{R}\}} \right\rangle_i \\
&= \frac{N}{\Omega} G_0^+(\mathbf{k}_2) G_0^+(\mathbf{k}_1) W_{k_2 k_1} \langle e^{-i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{R}_i} \rangle \quad (15)
\end{aligned}$$

In the first line we are pulling out terms that don't have any dependence on $\{\mathbf{R}\}$, and then pulling out the summation by including another average on the summation index. However, because the impurities are randomly distributed, there is no preferred coordinate and so for every value of i , the expectation value on the inside evaluates to the same thing. So we just drop the average over i . In the second line we are simply using (12), and the i is just a dummy index that can be anything from 1 to N .

To actually carry out this average, we recall that for any impurity ion, it has an equal probability of being anywhere in the volume. Imagine the entire volume is broken up into many infinitesimal volumes $d^3\mathbf{R}$: the probability that a certain ensemble has our impurity ion of choice within this volume is $d^3\mathbf{R}/\Omega$, since integrating this over the entire volume should give 1. If you'd like, you can think of it as grouping together every ensemble that has this particular impurity ion in some particular region, and all we're doing is then adding up every region's contribution, which by proxy adds up every ensemble's contribution. This yields:

$$\langle e^{-i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{R}_i} \rangle = \frac{1}{\Omega} \int d^3\mathbf{R} e^{-i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{R}} = \delta_{k_2 k_1} \quad (16)$$

The $k_1 = k_2$ case is simple: the integral just gives the volume and the expression yields 1. In the case they're not equal, the details are a bit messy so I'll describe it briefly. The idea is to treat this using Cartesian coordinates and splitting the exponential into a product of exponentials. Then convert them using Euler's identity, take one of the integrals and separate the cosine and sine terms into two different multivariable integrals, and note that the sine part will just vanish because of odd antisymmetry. The cosine part you can evaluate easily if you assume the volume is a box, and it becomes a one-dimensional integral. The result is of the form (say you picked the x integral) $\propto \sin(k_x L_x/2)$, which, since $k_x = 2\pi n_x/L_x$ due to quantization, is zero.

In the end we find:

$$\left\langle G_0^+(\mathbf{k}_2) \sum_{i=1}^N V_{k_2 k_1}(\mathbf{R}_i) G_0^+(\mathbf{k}_1) \right\rangle = \frac{N}{\Omega} G_0^+(\mathbf{k}_2) G_0^+(\mathbf{k}_1) W_{k_2 k_1} \delta_{k_2 k_1} \quad (17)$$

where we note that we do indeed see conservation of momentum as we suspected. Using the same trickery on the third term gives the following:

$$\begin{aligned}
\left\langle G_0^+(\mathbf{k}_2) \left[\sum_l \sum_{i=1}^N V_{k_2 l}(\mathbf{R}_i) G_0^+(l) V_{l k_1}(\mathbf{R}_i) \right] G_0^+(\mathbf{k}_1) \right\rangle \\
= \frac{N}{\Omega^2} G_0^+(\mathbf{k}_2) G_0^+(\mathbf{k}_1) \sum_l G_0^+(l) W_{k_2 l} W_{l k_1} \delta_{k_2 k_1} \quad (18)
\end{aligned}$$

Again we see that momentum conservation is enforced. The sum over l is better carried out as an integral, which corresponds to the substitution:

$$\sum_l \rightarrow \frac{\Omega}{(2\pi)^3} \int d^3\mathbf{l} \quad (19)$$

which can be done because k-space is dense in a large system and the prefactor is just the density of points in k-space. The way I think of the prefactor is originally, if we were in n-space, the sum just becomes a plain integral over n . This isn't that hard to see if you think of the limit where the integrand doesn't change appreciably over a unit volume. So once you make it an n-space integral, just change coordinates to a k-space integral and you get (19).

Now, (18) becomes:

$$\begin{aligned} & \left\langle G_0^+(\mathbf{k}_2) \left[\sum_l \sum_{i=1}^N V_{k_2 l}(\mathbf{R}_i) G_0^+(\mathbf{l}) V_{l k_1}(\mathbf{R}_i) \right] G_0^+(\mathbf{k}_1) \right\rangle \\ &= \frac{N}{\Omega} G_0^+(\mathbf{k}_2) G_0^+(\mathbf{k}_1) \int \frac{d^3\mathbf{l}}{(2\pi)^3} G_0^+(\mathbf{l}) W_{k_2 l} W_{l k_1} \delta_{k_2 k_1} \quad (20) \end{aligned}$$

That seems like a lot of work for nothing but the important point is the prefactor of N/Ω . This is particle density and is pretty important since we normally set it to something small, allowing for approximations. Note that each time we add another scattering off of the same impurity ion, there is one more potential to account for which adds another factor of Ω , while there is another sum over k-space we turn into an integral, which removes that extra factor of Ω as it did above. And since it's off the same impurity ion, there is only one sum over the ions which yields one factor of N . Therefore, scattering off of the same ion, regardless of the number of times it happens, will always have a propagator average proportional to N/Ω . We will see how this becomes important in a minute. I'll also make special note here of the fact that, though $\mathbf{l}$ does appear in the exponent of the exponentials in the potential terms, because the potentials are of the same impurity ion, the $\mathbf{l}$ terms cancel out in the exponent. This will be an important point for the next term we consider.

The fourth term in (14) goes similarly, but because the second sum over impurity ions excludes one ion, the factor actually becomes $N(N-1)$, and the V factors each contribute one factor of Ω in the denominator and one factor of W , with exponentials that are averaged over. This average again yields delta functions, and we find:

$$\begin{aligned} & G_0^+(\mathbf{k}_2) \left[\sum_l \sum_{j \neq i} V_{k_2 l}(\mathbf{R}_j) G_0^+(\mathbf{l}) \sum_{i=1}^N V_{l k_1}(\mathbf{R}_i) \right] G_0^+(\mathbf{k}_1) \\ &= G_0^+(\mathbf{k}_2) G_0^+(\mathbf{k}_1) \cdot \frac{N(N-1)}{\Omega^2} \sum_l G_0^+(\mathbf{l}) W_{k_2 l} W_{l k_1} \delta_{k_2 l} \delta_{l k_1} \\ &\approx \left(\frac{N}{\Omega} \right)^2 [G_0^+(\mathbf{k}_1)]^3 W_{k_1 k_2}^2 \delta_{k_2 k_1} \quad (21) \end{aligned}$$

Again we see a term conserving the initial momentum. Note that this time the intermediate momentum was forced to just become the initial momentum via the delta function. This happened because, unlike the previous term, the potentials were not located at the same $\mathbf{R}$, and so the $\mathbf{l}$ dependence did not cancel out as before. So essentially in carrying out this calculation, you end up performing (16) twice for both $\mathbf{R}_i$ and $\mathbf{R}_j$. So for terms like this, we do not have to convert the sum over $\mathbf{l}$ to an integral.

With these insights we can now write down diagrams for this series, the components of which I'll define in a moment.

$$\begin{aligned}
\langle \parallel \mathbf{k} \rangle &= \text{single line } \mathbf{k} + \text{bubble } \mathbf{k} + \text{bubble } \mathbf{k}, \mathbf{l} + \text{bubble } \mathbf{k} + \text{bubble } \mathbf{k}, \mathbf{l} + \dots \\
&+ \text{bubble } \mathbf{k}, \mathbf{l} + \dots + \text{bubble } \mathbf{k}, \mathbf{k} + \dots + \text{bubble } \mathbf{k}, \mathbf{p}, \mathbf{q}, \mathbf{p} + \dots \quad (22)
\end{aligned}$$

These diagrams carry largely the same meaning as before, with the bubbles representing interactions, the double line being a propagator, and the single line being a free propagator. Except this time, the interactions aren't strictly a probability or even a probability amplitude on its own, but rather a short-form for W_{lk} , where $\mathbf{l}$ is the wavenumber above the bubble, and $\mathbf{k}$ is the wavenumber below the bubble.

Note that in creating this diagram we have enforced all the δ -functions, so that $\mathbf{k}_1 = \mathbf{k}_2 = \mathbf{k}$. When there is some wavenumber other than $\mathbf{k}$, such as $\mathbf{l}$, a $\mathbf{k}$ -space integral over that wavenumber is understood. Though, this will not include the traditional factor

of Ω (see (19)), which is instead included in the other elements as follows: Each bubble counts for one factor of N/Ω unless a dashed line connects various bubbles, in which case all connected bubbles together only carry one factor of N/Ω . The dashed line is intended to mean all connected bubbles are interactions with the same impurity ion. As for why we do this, the explanation is a bit earlier in the document where we calculate the propagators for multiple ions and separately consider scattering off of the same ion versus different ions.

In the book every element carries a factor of $\pm i$ but you can express this series even without it and it works out. Though, conventionally the right thing to do is to include those factors.

Anyway, this is still pretty complicated to do, so we make approximations. We assume that only the terms that contain single or double scattering by the same impurity are relevant, which is basically just to say the added particle goes through the system without really getting stuck in the middle. Under this constraint, terms like the last term on the first line of (22) are ignored (since it contains three scatterings off the same impurity ion), and the last term of (22) is also ignored (since it contains a scattering within a double scattering, which we also ignore). The only terms we count are the first, second and third terms of (22), and blocks you build with combinations of them. If you'd like, think of the second and third terms as the monkey and the lion, and recall the Feynman diagram on page 8, and find the result of it on page 16. This allows us to write the same result for this system:

$$\parallel \uparrow = \left[\uparrow^{-1} - \bigcirc \Sigma \right]^{-1} \quad (23)$$

In Mattuck it's actually as $+$ sign because of the convention he uses. Essentially, all the convention does is to slap on a factor of i to the propagators and a $-i$ to the bubbles. So you can imagine over here it just turns the $(-)$ to a $(+)$. But either way we can now put this in a functional form like so:

$$\langle G(\mathbf{k}, \omega) \rangle = \frac{1}{\omega - \epsilon_k + i\delta - \Sigma(\mathbf{k}, \omega)} \quad , \quad \Sigma(\mathbf{k}, \omega) = \frac{N}{\Omega} W_{kk} + \frac{N}{\Omega} \int \frac{d^3\mathbf{l}}{(2\pi)^3} \frac{|W_{kl}|^2}{\omega - \epsilon_l + i\delta} \quad (24)$$

where we have used the fact that $W_{kl} = W_{lk}^\dagger$ as we noted before. Using our previous theorem of the energies of the system corresponding to the poles of the propagator we find that up to a factor of $\hbar$ the energy is given by ω that solves:

$$\omega - \epsilon_k - \Sigma(\mathbf{k}, \omega) + i\delta = 0 \quad (25)$$

Really the only non-triviality in (24) is in this new Σ term, so our goal is to try to express this in a more informative way. But it's fairly clear that in its current form it isn't really feasible to find an expression for ω , for it's hiding inside an integral we can't exactly evaluate. Something we can do to alleviate this is to use a first order approximation, by plugging in the zeroth-order approximation to ω into Σ and then solve for ω . The zeroth-order approximation is of course none other than ϵ_k .

Seeing (3), we may also want to write our solution as a real and imaginary part, which we can do by multiplying the numerator and denominator by $\omega - \epsilon_l - i\delta$. This makes

the denominator real evidently, while cleanly introducing a real and imaginary part on the numerator. Using the aforementioned approximation along with this step gives the following:

$$\begin{aligned}\text{Re } \Sigma(\mathbf{k}, \epsilon_k) &= \frac{N}{\Omega} W_{kk} + \lim_{\delta \rightarrow 0} \frac{N}{\Omega} \int \frac{d^3 \mathbf{l}}{(2\pi)^3} \frac{|W_{lk}|^2 (\epsilon_k - \epsilon_l)}{(\epsilon_k - \epsilon_l)^2 + \delta^2} \\ &= \frac{N}{\Omega} W_{kk} + \frac{N}{\Omega} P \int \frac{d^3 \mathbf{l}}{(2\pi)^3} \frac{|W_{lk}|^2}{\epsilon_k - \epsilon_l}\end{aligned}\quad (26)$$

$$\begin{aligned}\text{Im } \Sigma(\mathbf{k}, \epsilon_k) &= -\lim_{\delta \rightarrow 0} \frac{N}{\Omega} \int \frac{d^3 \mathbf{l}}{(2\pi)^3} \frac{|W_{lk}|^2 \delta}{(\epsilon_k - \epsilon_l)^2 + \delta^2} \\ &= -\pi \frac{N}{\Omega} \int \frac{d^3 \mathbf{l}}{(2\pi)^3} |W_{lk}|^2 \delta(\epsilon_k - \epsilon_l)\end{aligned}\quad (27)$$

where in (26) P stands for principal part which is necessary since the integral isn't exactly defined properly throughout the integration range. We have put in the limit for δ now that we've actually reached the end of our calculations, and indeed it does give us something nontrivial particularly in (27), where it turns the expression into a delta function. It's pretty easy to see it satisfies the condition of blowing up when $\epsilon_k - \epsilon_l = 0$, while being 0 for finite differences. The normalization condition is also not very hard to check.

Looking at (3) and (25) and noting that $\epsilon'_k = \epsilon_k + \text{Re } \Sigma$, $\tau_k^{-1} = -\text{Im } \Sigma$, we find:

$$\epsilon'_k = \epsilon_k + \frac{N}{\Omega} W_{kk} + \frac{N}{\Omega} P \int \frac{d^3 \mathbf{l}}{(2\pi)^3} \frac{|W_{lk}|^2}{\epsilon_k - \epsilon_l}\quad (28)$$

$$\tau_k^{-1} = \pi \frac{N}{\Omega} \int \frac{d^3 \mathbf{l}}{(2\pi)^3} |W_{lk}|^2 \delta(\epsilon_k - \epsilon_l)\quad (29)$$

Note that (28)'s second term looks a lot like second order perturbation theory, which tells us we've done something right here. The term (29) is a manifestation of Fermi's golden rule in the following sense. The $|W_{lk}|^2$ term is the usual transition amplitude term, and the delta function is the density of states. Integrating over all the final states, this represents roughly the probability that the original state jumps to any other state, per unit time. So when you flip it over, it becomes the number of time units it takes for the particle to go to a different state, which is precisely what the lifetime is. Very nice stuff. Let's do some problems now.

Problem 1

Find the wavefunctions and energies of an infinite square well stretching from $x = 0$ to $x = a$. Find the free propagator and its Fourier transform.

Solution

To find the wavefunctions and energies go have a look at Griffiths Section 2.2. The basic idea is to use the time-independent Schrödinger equation while enforcing continuity of the wavefunction, and then normalize. Within the well it's a free particle, and outside the well the wavefunction is zero. The answer changes somewhat in form based on where you choose the bounds of the well to be, but for this particular choice (which happens to align with Griffith's, and is the simplest) the answer goes like so:

$$\phi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right), \quad \epsilon_n = \frac{n^2\pi^2}{2ma^2} \quad (30)$$

where as usual we've set $\hbar = 1$. These functions are orthonormal just like the free particle case we've already dealt with, and the time-dependent Schrödinger tacks on the exact same time dependence, but with a new energy. So, essentially, every result we've already found applies, except that we replace the free particle energy with the infinite square well energy. So we find:

$$G_0^+(n, t_2 - t_1) = \begin{cases} -i\theta_{t_2-t_1} e^{-i\epsilon_n(t_2-t_1)}, & t_2 \neq t_1 \\ 0, & \text{otherwise} \end{cases} \quad (31)$$

$$G_0^+(n, \omega) = \frac{1}{\omega - \epsilon_n + i\delta} \quad (32)$$

with the energies given in (30).

Problem 2

Consider a perturbation to the infinite square well ($U(x)$) from above of the form $V(x) = B(p^2/2m + U(x))^3$. Calculate the transition amplitude from $\phi_n(x)$ to $\phi_m(x)$ under the influence of this perturbation.

Solution

The transition amplitude is just $V_{mn} = \langle \phi_m | V | \phi_n \rangle = B(k_n^2/2m)^3 \delta_{mn}$ where $k_n = n\pi/a$. Only the kinetic term matters here since the $U(x)$ term is zero inside the well (so it really is just the kinetic term) whereas outside the well is not really a point of interest since the wavefunction has to be zero there anyway (and the perturbation doesn't change that).

Problem 3

Write a diagram series for the propagator and evaluate it.

Solution

I mean, this is just the strong interaction sum. Feynman diagrams take forever so I'm not going to write it out but it's just what you see in the middle of page 3. The answer is just (6) but Mk^2 becomes the new transition amplitude we found in the previous problem. You get something like this:

$$G^+(n, \omega) = \frac{1}{(G_0^+)^{-1} - V_{nn}} = \frac{1}{\omega - (\epsilon_n + B(k_n^2/2m)^3) + i\delta} \quad (33)$$

where I've defined the meanings of the intermediate terms previously.

Problem 4

Find the quasi particle energy dispersion law and lifetime.

Solution

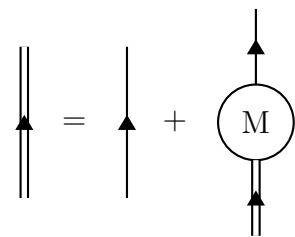
You can just read this one off based on (3), and you find:

$$\epsilon'_n = \epsilon_n + B \left(\frac{k_n^2}{2m} \right)^3 \quad (34)$$

$$\tau_n = \infty \quad (35)$$

Problem 5

Recall the following Feynman diagram:



$$\text{Diagram} = \text{Diagram} + \text{Diagram} \quad (36)$$

Show that, when written in $(\mathbf{k}, t)$ space, it takes the following form:

$$G^+(\mathbf{k}, t - t') = G_0^+(\mathbf{k}, t - t') + \int_{-\infty}^{\infty} dt'' G_0^+(\mathbf{k}, t - t'') V(\mathbf{k}) G^+(\mathbf{k}, t'' - t') \quad (37)$$

Solution

In $(\mathbf{k}, t)$ space the Feynman diagram looks like so:

The diagram shows an equation labeled (38). On the left-hand side (LHS), there are two parallel vertical lines with upward-pointing arrows. The top line is labeled $\mathbf{k}, t$ and the bottom line is labeled $\mathbf{k}, t'$. This is followed by an equals sign. On the right-hand side (RHS), there is a single vertical line with an upward arrow labeled $\mathbf{k}, t$ and $\mathbf{k}, t'$ at its base. This is followed by a plus sign and a circular vertex labeled V . Two lines enter the vertex from below: one labeled $\mathbf{k}, t'$ and the other labeled t'' . A single line exits the vertex from above, labeled $\mathbf{k}, t$.

At this point it's fairly obvious, I would think. The LHS and the first term on the RHS are pretty clear, at least, they are just the definitions of what the diagram means. The second term in (38) corresponds to the integral in (37) in the same way we've discussed before: the second diagram in (38) has total propagation from t' in the wavefunction with wavenumber $\mathbf{k}$ up to some interaction modeled by V at some intermediate time t'' , and then free propagation from t'' to time t in the same wavefunction state. Of course, the time t'' can occur any time in the interval (t', t) , and so we integrate over t'' . This is precisely what you see in (37).

5 Fermions (Ch. 4)

This chapter finally gets into a many body system, and particularly a Fermi system. To be clear we do not do much with interacting particles but fermions do follow the Pauli exclusion principle so a system with many of them is not exactly trivial.

I won't go through the theory of a Fermi system in its ground state but what we need to know is minimal: the idea of a Fermi sea, the Fermi energy level ϵ_F , the corresponding Fermi momentum k_F , and the Fermi surface. The ground state consists of all the fermions below the Fermi surface and excited states have fermions above the Fermi surface with "holes" underneath the Fermi surface. Note that "holes" by definition (at least here) are points without a fermion where the ground state would normally have one there.

The details on how things look like in k -space change depending on the external potential but the discussion in this chapter doesn't really require worrying about this, we'll only think about a free system.

It's annoying to represent every fermion in the system so it's helpful to think of the system in a "particle-hole" way, which is to take the ground state as "empty", and every excited state is represented by noting down the excited fermion(s) and the holes that formed in the places they came from. These kinds of particles are stylistically called *particles* with an italicized p , and can only exist above the Fermi surface while their corresponding holes can only exist below the Fermi surface. Most of our discussion will be using *p* particles.

Holes can be thought of as particles in their own way, in the following sense. You can roughly think of a hole as having the negative of the energy of a particle since it's like "removing" a particle. If you plug that into the time evolution of a state you find for a hole $\psi_{\text{hole}}(t) = \phi_k e^{-i(-\epsilon_k)t}$, and if you identify the minus sign with time instead you can roughly think of it as a particle moving backwards in time. This justifies some of our earlier remarks with holes in Feynman diagrams. This is a purely theoretical and strictly unphysical description made for convenience.

Many body Fermi systems are a little bit of a mess and can use a similar trick. Clearly the total wavefunction must be antisymmetric upon exchange of any two fermions so

what we must do in a system of N fermions is to take the N states the fermions are in $\phi_{k_1}, \dots, \phi_{k_N}$, and their N locations $\mathbf{r}_1, \dots, \mathbf{r}_N$ and account for every single state in every single location, and do this in such a way that it is antisymmetric upon exchanging positions or states. This is accomplished by the Slater determinant:

$$\phi_{k_1, \dots, k_N}(\mathbf{r}_1, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \phi_{k_1}(\mathbf{r}_1) & \cdots & \phi_{k_1}(\mathbf{r}_N) \\ \vdots & & \vdots \\ \phi_{k_N}(\mathbf{r}_1) & \cdots & \phi_{k_N}(\mathbf{r}_N) \end{bmatrix} \quad (39)$$

That this satisfies our requirements is simple to reason if you recall the determinant form using the Levi-Civita tensor: exchanging positions means exchanging columns and exchanging states means exchanging rows, which both swap indices in the tensor which is antisymmetric by definition, and so everything works out.

When interactions are involved the general state becomes a linear superposition of states of the form (39) for different choices of k_i . In such a situation using something like (39) is very tiresome and so it helps to use a ket notation that is easier to look at and write down.

One way to do this is to take (39) and note down how many fermions it has in state ϕ_{p_1}, ϕ_{p_2} , etc. where the k_i 's were arbitrary but now we take the p_i 's to be ordered by energy, and then use this knowledge to write (39) as a series of ones and zeroes (since two fermions can't be in the same state). So for instance a ground state looks like $|1, 1, 1, 1, 0, 0, \dots\rangle$ and an excited state looks like $|1, 1, 0, 1, 1, 0, \dots\rangle$.

Now, the particle-hole trick can be used here again, noting the ground state as a "vacuum" state (Fermi vacuum) $|0\rangle$, and simply noting down the changes, for instance like $|1_h^3, 1_p^8\rangle$ where the subscripts i are short for p_i , the h superscript stands for hole and the p superscript stands for a particle. So in this state a fermion jumped from ϕ_{p_3} to ϕ_{p_8} from the ground state.

Here we introduce the creation and annihilation operators similar to the ladder operators, which satisfy the following relations:

$$c_i |n_1, \dots, n_i, \dots\rangle = n_i |n_1, \dots, n_i - 1, \dots\rangle \quad (40)$$

$$c_i^\dagger |n_1, \dots, n_i, \dots\rangle = (1 - n_i) |n_1, \dots, n_i + 1, \dots\rangle \quad (41)$$

where, since n_i can only be zero or one, the prefactors essentially force this condition. That is, you can't add two fermions to the same state and you can't remove a fermion from a state where there are no fermions. Thinking of this in the lens of particle-holes, there are now four operators, two for particles, and two for holes which satisfy:

$$c_i = \theta_{k_i - k_F} a_i + \theta_{k_F - k_i} b_i^\dagger \quad (42)$$

$$c_i^\dagger = \theta_{k_i - k_F} a_i^\dagger + \theta_{k_F - k_i} b_i \quad (43)$$

where the θ -functions essentially say that above the Fermi level the creation and annihilation operators are the same as the particle creation and annihilation operators, whereas below the Fermi level "annihilating" a particle means "creating" a hole and "creating" a particle means "annihilating" a hole.

Now, the ket notation is just a different way to write things so the following equality must hold:

$$\langle 00\dots 1_i 0\dots | \hat{O} | 0\dots 1_j 0\dots \rangle = \langle \phi_i | \hat{O} | \phi_j \rangle \equiv O_{ij} \quad (44)$$

Assuming we already have all the O_{ij} this is not that hard to guess: you just want to write it as $\sum_{ij} O_{ij} c_i^\dagger c_j$ and obviously that satisfies (44).

One of the advantages of this notation is that the many body Hamiltonian with mutually interacting particles, we can write:

$$\hat{H} = \underbrace{\sum_i \frac{\hat{p}_i^2}{2m}}_{\hat{H}_0} + \underbrace{\frac{1}{2} \sum_{ij} V(\mathbf{r}_i - \mathbf{r}_j)}_{\hat{H}_1} \quad (45)$$

↓

$$\hat{H}_0 = \sum_{k < k_F} \epsilon_k a_k^\dagger a_k + \sum_{k < k_F} \epsilon_k b_k b_k^\dagger, \quad \epsilon_k = k^2/2m \quad (46)$$

$$\begin{aligned} \hat{H}_1 = \frac{1}{2} \sum_{k,l,m,n > k_F} V_{klmn} a_l^\dagger a_k^\dagger a_m a_n + \frac{1}{2} \sum_{\substack{k,l,m > k_F \\ n < k_F}} V_{klmn} a_l^\dagger a_k^\dagger a_m b_n^\dagger + \dots \\ + \frac{1}{2} \sum_{k,l,m,n < k_F} V_{klmn} b_l b_k b_m^\dagger b_n^\dagger \end{aligned} \quad (47)$$

with:

$$V_{klmn} = \int d^3\mathbf{r} \int d^3\mathbf{r}' \phi_k^*(\mathbf{r}) \phi_l^*(\mathbf{r}') V(|\mathbf{r} - \mathbf{r}'|) \phi_m(\mathbf{r}) \phi_n(\mathbf{r}') \quad (48)$$

Equation (47) will come in handy when we deal with Feynman diagrams, and so will (48). Note that (48) is symmetric upon exchanging both the first two and last two indices, i.e. $V_{klmn} = V_{lnmk}$.

This is enough background to get started in earnest. Since we have this idea that the hole is like a particle going back in time we can think of assigning to it the advanced propagator with the following interpretation: $iG(k_2, k_1, t_2 - t_1)_{t_2 \leq t_1} \equiv iG^-(k_2, k_1, t_2 - t_1)$ is equal to (minus) the probability that if you add a hole in ϕ_{k_2} at time t_2 in the ground state of the system, then there will be a hole in the ground state of the system at time t_1 in state ϕ_{k_1} . The minus sign is some fermion stuff. For positive time arguments the advanced propagator is defined as zero, but this is not true for $t_2 = t_1$ as we'll see in a moment.

The argument for defining the advanced free propagator in this sense is not difficult: it is the same argument as for the particle, except you have to take into account the minus sign and the time coordinates for which it is nonzero. Also note that this time the

"final" state is k_1 and the "initial" state is k_2 : this is why the exponent stays the same. This yields:

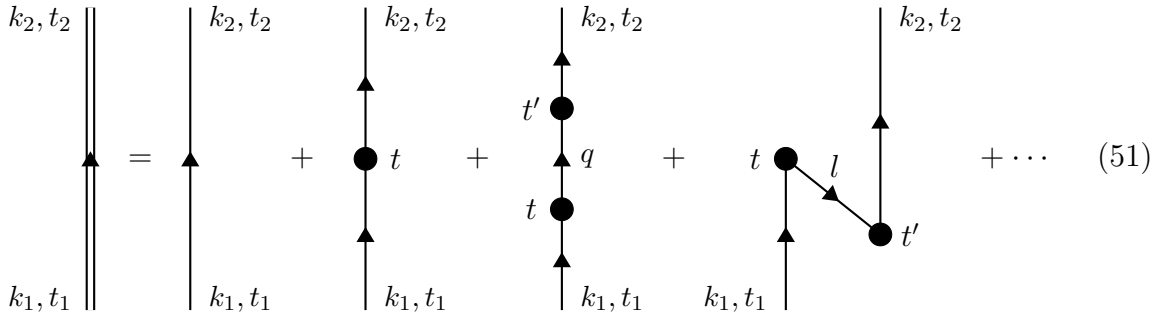
$$G_0^-(k, t_2 - t_1) = \begin{cases} i\theta_{t_1-t_2} e^{-i\epsilon_k(t_2-t_1)}, & t_2 \neq t_1, \epsilon_k < \epsilon_F \\ i, & t_2 = t_1 \end{cases} \quad (49)$$

The definition at $t_2 = t_1$ is... not easy to explain now so we'll (potentially) take it up later. This has Fourier transform:

$$G_0^-(k, \omega) = \frac{1}{\omega - \epsilon_k - i\delta} \quad (50)$$

where the sign on $i\delta$ differs from (2) because the θ function restricts the integral to the other half, and so to regularize the integral you end up having to introduce δ with the opposite sign in the exponent.

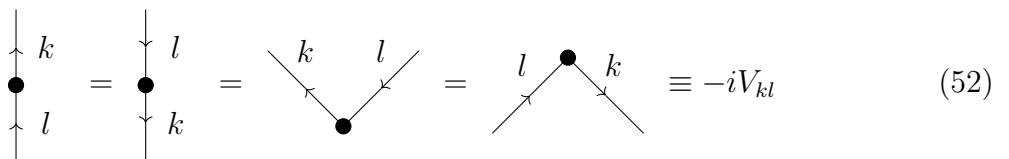
Let's now consider a system where the particles aren't actually interacting, but they do sit in an external potential. The idea is to use the intuitive picture of a Feynman diagram and run the machine backward: a particle comes in with momentum k_1 , and a particle goes out with momentum k_2 , and so the diagram series ends up looking like so:



$$\begin{array}{c} k_2, t_2 \\ \parallel \\ \uparrow \\ k_1, t_1 \end{array} = \begin{array}{c} k_2, t_2 \\ \uparrow \\ k_1, t_1 \end{array} + \begin{array}{c} k_2, t_2 \\ \uparrow \bullet t \\ \uparrow \\ k_1, t_1 \end{array} + \begin{array}{c} k_2, t_2 \\ \uparrow \bullet t' \\ \uparrow q \\ \uparrow \bullet t \\ \uparrow \\ k_1, t_1 \end{array} + \begin{array}{c} k_2, t_2 \\ \uparrow \bullet t \\ \swarrow l \\ \bullet t' \\ \uparrow \\ k_1, t_1 \end{array} + \dots \quad (51)$$

where the ellipses include terms like the third one but with further intermediate interactions, or terms like the fourth but with more holes kicked up in the middle, and of course combinations of all of them. As a matter of convention the the lines themselves are the usual propagators but with a factor of i , whereas the black circles are $-iV_{kl}$ where k is the outgoing momentum and l is the incoming momentum. You can crudely think of, say, the left circle in the fourth diagram as k_1 coming in and l "going out".

Of course this is not what's physically going on but if you think of it that way the signs work out. As for why, you can check for yourself that it works out using the definitions of the wavefunctions of the hole and particle and then taking the inner product. In fact all of the following are equivalent using this idea:



$$\begin{array}{c} \uparrow k \\ \bullet \\ \uparrow l \end{array} = \begin{array}{c} \downarrow l \\ \bullet \\ \downarrow k \end{array} = \begin{array}{c} \swarrow k \\ \bullet \\ \swarrow l \end{array} = \begin{array}{c} \swarrow l \\ \bullet \\ \swarrow k \end{array} \equiv -iV_{kl} \quad (52)$$

The sum (51) is translated into functional form in exactly the same way as we've seen thus far, where each portion of the diagram is treated as independent and just multiplied,

and then intermediate times and states are integrated over their respective domains, and then you can change it to Fourier space fairly simply as well.

Note the interpretation of terms like the last one in (51): a particle-hole pair form at t' , and the hole annihilates with the incoming particle, while the formed particle exits in state k_2 . It's worth mentioning that there is no way to know "which" fermion is coming out. Nevertheless, discussions like this are made much more convenient if we think of them as distinguishable.

An example where (51) has a simple solution is if only V_{mk} and V_{km} are relevant, where $m < k_F$ and $k = k_1 = k_2$. In this case the second and third terms in (51) are negligible, while the fourth term and combinations involving the middle portion will contribute. So once again the slanted line plays the role of the monkey and the result is approximately:

$$\text{Diagram: a vertical double line with an upward arrow} \approx \left[\text{Diagram: a vertical line with an upward arrow} - \text{Diagram: a loop with two vertices and a line labeled } m \text{ between them} \right]^{-1} \quad (53)$$

which in functional form becomes;

$$G^+(k, \omega) = \frac{1}{[G_0^+(k, \omega)]^{-1} - V_{km} V_{mk} G_0^-(m, \omega)} = \frac{1}{\omega - \epsilon_k + i\delta - \frac{|V_{km}|^2}{\omega - \epsilon_k - i\delta}} \quad (54)$$

If you're solving for the energies, you can solve for the poles, just setting δ to zero. This yields a quadratic equation with solutions:

$$\omega = \frac{\epsilon_k + \epsilon_m}{2} \pm \frac{1}{2} \sqrt{(\epsilon_k - \epsilon_m)^2 + 4|V_{km}|^2} \quad (55)$$

Equation (55) is a general result for two level systems, with the plus solution corresponding to particles and the minus solution corresponding to holes. In the limit of $V_{km} = 0$ this is fairly clear.

In a situation where the particles can actually interact, terms like (48) are needed. Diagrammatically:

$$\text{Diagram: a wavy line connecting two vertices, each with two external lines labeled } k, m \text{ and } l, n \text{ respectively} \equiv -i \frac{1}{2} V_{klmn} \quad (56)$$

The various terms you see in (52) can take the place of any junction. Also note that the interactions we consider conserve momentum so, strictly speaking, if $\mathbf{k} = \mathbf{m} + \mathbf{q}$, then $\mathbf{l} = \mathbf{n} - \mathbf{q}$. In fact as far as the interaction amplitude goes even m, n are redundant.

Diagrams using (56) do not strictly have to conserve energy at any intermediate time, meaning these processes are *virtual* rather than real. In the same vein the Pauli exclusion principle can also be violated which is actually true even in the last term in (51). However, again, these are virtual processes and further all diagrams violating the Pauli principle cancel out.

This is mind-numbingly boring so I am going to just briefly cover the rest. Essentially, when you try to write (56) with all possible combinations of (52), not all of them contribute. This is because the "proper" way of thinking of an interaction here is to think of removing two particles/holes and adding two particles/holes. In the operator formalism this doesn't happen in whatever order, and in fact the order is dictated by (47): the amplitude is the potential term, and, for instance, the first term says that in an interaction between two particles n is removed first, then m , then k is added, and then n is added. Every interaction starts at the ground state after which the operators are applied, and so some of these terms just become zero.

Of the ones that don't vanish, there are only two unique ones (up to a 180 degree rotation which does nothing). If you connect the hole propagators which shouldn't "come out" of the interaction for a first-order interaction, you end up getting exactly the open oyster and also a "fermion loop". Physically the fermion loop corresponds to instantaneously creating and annihilating a hole, and the particle continuing on, while the open oyster has an incoming particle that instantaneously creates a hole that it instantaneously fills, kicking up the particle that was there, which then exits.

The hole that's created can take on any momentum below the Fermi surface so that must be summed over. The fermion loop itself has a factor of minus 1 for... reasons. And from (49) you can see iG_0^- when the time argument is zero (i.e. instantaneous propagation) is just minus 1. So you end up finding that if you think of the fermion loop as the monkey and that's the only important interaction you find:

$$G^+(\mathbf{k}, \omega) = \frac{1}{\omega - \epsilon_k - \sum_{l < k_F} V_{klkl} + i\delta} \quad (57)$$

from which you can find the usual quasi particle stuff. This is the Hartree approximation. Hartree-Fock is if you treat it like the monkey and the lion but the lion is the open oyster, and that gives you:

$$G^+(\mathbf{k}, \omega) = \frac{1}{\omega - \epsilon_k - \sum_{l < k_F} (V_{klkl} - V_{lkkl}) + i\delta} \quad (58)$$

And then sections 4.8 and 4.9 just apply the energy solutions from the poles of (57) and (58) using a definite form for V , performing the integrals to find V_{klmn} , and then performing the sum over the states using a k -space integral. He skips every step anyway so not that big of a deal. In 4.8 he does this for the Yukawa potential, and in 4.9 he does this for a random electron gas, and then introducing the RPA which is only schematically anyway. You can also find an "effective mass" if you set the resulting energy to $k^2/2m^*$ and ignore constant terms.

Okay now let's get to solving problems.

Problem 1

In a system of free particles a hole is formed in the single-particle state $\phi_k(\mathbf{r}) = \Omega^{-1/2} e^{i\mathbf{k}\cdot\mathbf{r}}$. What is the momentum of the hole?

Solution

$-\mathbf{k}$. The idea of a hole having attributes like energy and momentum only makes sense in the particle-hole formalism, and in that case you note down that you have a hole in $\mathbf{k}$ and a *particle* in some state above the Fermi surface. Now, physically the system has the momentum of Fermi vacuum plus the hole momentum plus the *particle* momentum, but if you go back to the original picture the momentum of the system is Fermi vacuum minus the particle that was lost at $\mathbf{k}$ plus the momentum of wherever it ended up. So it's clear when matching the two situations that the hole momentum is $-\mathbf{k}$.

Problem 2

- (a) Evaluate $c_3^\dagger c_6^\dagger c_4 c_6^\dagger c_3 |1111100\dots\rangle$
- (b) Write $|1101100100\dots\rangle$ in particle-hole notation.
- (c) Find $\sum_k \epsilon_k c_k^\dagger c_k |1111100\dots\rangle$

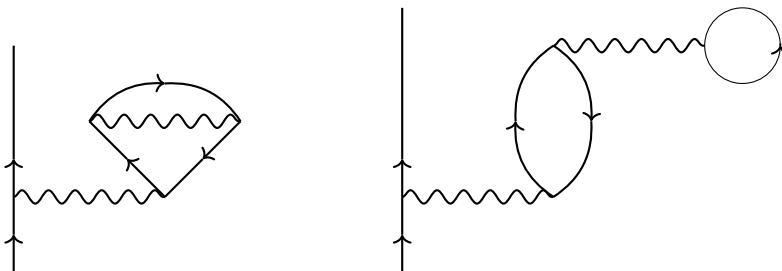
Solution

- (a) This is evidently zero since you try to add twice to ϕ_{p6} . See (41).
- (b) This is already in the text, $|1_3^h, 1_8^p\rangle$.
- (c) This is just the Hamiltonian acting on it so you expect to pull out the energy. To see why it's the Hamiltonian, look at (44) and the line after.

Now doing it properly, note that terms beyond $k = 5$ will just give zero, see (40). Between $1 \leq k \leq 5$ a particle will be removed then added so that the state is unchanged but will pick up a prefactor of ϵ_k . So this just turns into the sum of all the energies for which a fermion exists, i.e. $\sum_{k=1}^5 \epsilon_k$. Finally you get $(\sum_{k=1}^5 \epsilon_k) |1111100\dots\rangle$ which is exactly what we expected.

Problem 3

Explain why the following Feynman diagrams are not allowable on the grounds of momentum conservation.

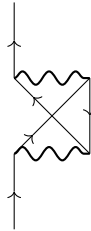


Solution

As the problem states the issue is momentum conservation. Note that the particle never interacts with the system again, so if it is to leave with the same momentum, the interaction could not have exchanged any momentum. But then the entire idea of the particle-hole pair is nonsense: if no momentum was imparted, the particle could not have gone to a different momentum state, so no hole could have formed in its stead. Therefore these diagrams are not allowed.

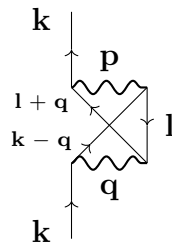
Problem 4

Label the following diagram explicitly to show momentum conservation and show that the scattering processes at each interaction are virtual.



Solution

The momentum coming in should be the momentum coming out, while interactions are capable of transferring momentum. Recalling what we said in the previous problem, the momentum exchanged must be nontrivial. So you get something like this (I'm omitting ω 's):



where $\mathbf{p} = \mathbf{l} + \mathbf{q} - \mathbf{k}$ (I'm assuming the momentum flows to the right by default). In the first interaction the energy coming in is $k^2 + l^2$ and the energy going out is $(\mathbf{k} - \mathbf{q})^2 + (\mathbf{l} + \mathbf{q})^2$, whereas in the next interaction it's the opposite. These interactions don't conserve energy so they're virtual.

Problem 5

Translate Figure 1 to its functional form.

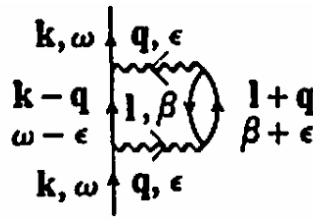


Figure 1: Problem 5 Feynman Diagram

Solution

The functional form cannot depend on anything but $\mathbf{k}, \omega$: everything else must be summed over or integrated, which gives something like this, just reading off every line:

$$G^+(\mathbf{k}, \omega) = \int \frac{d\beta}{2\pi} \int \frac{d\epsilon}{2\pi} \sum_{l < k_F} \sum_{\substack{\mathbf{q} \text{ s.t.} \\ |\mathbf{q} + \mathbf{l}| > k_F}} [G_0^+(\mathbf{k}, \omega)]^2 [V_q]^2 \{ G_0^+(\mathbf{k} - \mathbf{q}, \omega - \epsilon) G_0^-(\mathbf{l}, \beta) G_0^+(\mathbf{l} + \mathbf{q}, \beta + \epsilon) \}$$

Note that the intermediate frequencies are continuous so they must be integrated, and with a factor of 2π because that's our convention for the Fourier transform. As for the limits of the sum, it is clear that $\mathbf{l}$ must have magnitude below the Fermi level because it describes a hole, while $\mathbf{l} + \mathbf{q}$ is the momentum of a *p*article so it must have magnitude over the Fermi sea. As long as these conditions hold, $\mathbf{l}, \mathbf{q}$ can be anything.

Also I've omitted factors of i but you can easily check they actually do just vanish. To see this note that there are two interactions and five propagators which give seven factors of i , giving $-i$. The minus signs from the interactions cancel and we pick up a minus sign from there being a fermion loop (the pair bubble). So overall it's a factor of i , which you can just cancel with the i on the LHS.

Problem 6

Show the identity in Figure 2.

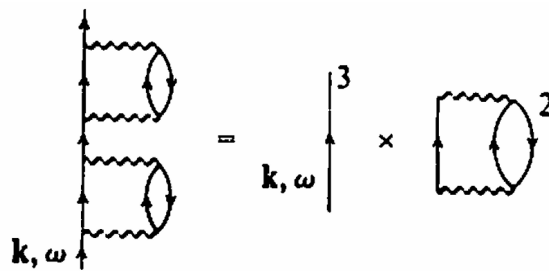


Figure 2: Problem 7

Solution

Interactions conserve momentum (not necessarily energy), which means at every time slice if you add up all the momenta you will get $\mathbf{k}$. Of course for this to make sense you have to know the momentum of a hole as we found in the first problem. Anyway, then, it is clear the middle propagator between the double interaction terms must have wavenumber $\mathbf{k}$, which means you can pull it out since there's no sum over it. Then it's pretty trivial: you just multiply the individual elements like we always do. Of course we can't break it down any further because the individual elements on the term on the right have sums over their wavenumbers.

Problem 7

Take Hartree-Fock and add on another relevant term, particularly the double scattering term given in Figure 1. Find the approximate propagator using partial summation.

Solution

This just turns into permutations of three possible interactions which again has precisely the same idea as two strong interactions. The sum just becomes:

$$\text{Diagram 1} \approx \left[\text{Diagram 2} - \text{Diagram 3} - \text{Diagram 4} - \text{Diagram 5} \right]^{-1} \quad (59)$$

And then you plug in the functional forms, like in Mattuck eq. (4.77) but including the solution to Problem 5.

Problem 8

Show that (60) and (61) converge to (62) in the limit $V_0 a = e^2$, $a \rightarrow \infty$.

$$\epsilon'_k = \frac{k^2}{2m} + \frac{2V_0 a^3 k_F^3}{3\pi} - \frac{V_0}{2\pi} [F(ka + k_F a) - F(ka - k_F a)] \quad (60)$$

$$F(z) = \frac{1}{2ka} [1 + z^2] [\ln(1 + z^2) - 1] - z \ln(1 + z^2) + 2z - 2 \tan^{-1} z \quad (61)$$

$$\epsilon'_k = \frac{k^2}{2m} - \frac{e^2 k_F}{2\pi} \left[2 + \frac{(k_F^2 - k^2)}{kk_F} \ln \left| \frac{k + k_F}{k - k_F} \right| \right] \quad (62)$$

Solution

This is going to take forever so I'll just walk through it briefly. Essentially plug in $V_0 = e^2/a$. Take the limit of (61) when the magnitude of z is very large. This simplifies the logarithm and the tangent inverse. Then plug in the value for z and keep terms of order a and higher (highest order will be $a \ln a$ which is easily seen from (61)), the rest are not relevant. Then plug into (60) to find (62). In my notes for whatever reason the second term in (60) doesn't go away but I guess it's ignorable because it's a constant.

Problem 9

Take a system of N non-interacting fermions, with an external potential that satisfies $V_{kl} = A f_k f_l$ for all $\mathbf{k}, \mathbf{l}$. Find $G^+(\mathbf{q}, \mathbf{p}, \omega)$ where $q, p > k_F$ exactly.

Solution

The exact diagram series is (51) (more terms shown in Mattuck (4.34)). Ordinarily the innards are all correlated and it's a mess, but the advantage of the form of the potential here is that all the inside portions can be separated so that you can treat them as independent. As an example let's look at the following diagram from (51).



This becomes $G_0^+(\mathbf{k}_1)G_0^+(\mathbf{k}_2)A^2 f_{k_1} f_{k_2} \sum_q f_q^2 G_0^+(\mathbf{q})$. It's not hard to reason that every single diagram except the free propagator will include the prefactor except for the A^2 term, which will depend on the number of interactions. What we have found here is that this middle portion, which is normally an ugly sum correlated with the outer propagators as well, has a nice separated form that you can call "the monkey". In a similar fashion you can call the intermediate hole term "the lion". And then it's just the monkey and the lion we've seen all along. To this end, define:

$$\alpha = A \sum_{q > k_F} f_q^2 G_0^+(\mathbf{q}) \quad (63)$$

$$\beta = A \sum_{q < k_F} f_q^2 G_0^-(\mathbf{q}) \quad (64)$$

To think of them diagrammatically, it's easiest to say the propagator line plus one dot corresponds to α or β . If you do it this way then one of the dots can just be taken to encode $Af_{k_1}f_{k_2}$. With this it's possible to express the diagram series in the usual monkey and lion formalism, but unlike before it's actually possible to go to a different momentum state from the beginning, so the sum is a little different. In particular the first free propagator term in (51) becomes a Kronecker delta, while the second term is the first term in the power series. Essentially you pull out the second term (which, based on our diagram definitions is two free propagators on either side multiplied with $Af_{k_1}f_{k_2}$) from the rest of the series, and voilà, up pops the monkey and lion series that we know and love. Putting these ideas together we find (omitting ω as usual):

$$G^+(\mathbf{q}, \mathbf{p}) = G_0^+(\mathbf{p})\delta_{qp} + \frac{G_0^+(\mathbf{q})G_0^+(\mathbf{p})Af_qf_p}{1 - (\alpha + \beta)} \quad (65)$$

6 Vacuum (Ch. 5)

The primary object of interest in this chapter is the vacuum propagator $R(t)$, defined as the transition amplitude from Fermi vacuum (at time 0) to Fermi vacuum (at time t), alternatively called the no-particle propagator. Of particular interest is the following familiar looking relation:

$$E_0 = W_0 + \lim_{t \rightarrow \infty(1-i\eta)} i \frac{d}{dt} \ln R(t) \quad (66)$$

Equation (66) yields the ground state energy, where W_0 is the ground state energy of the free system. The diagrams associated with vacuum are somewhat more complicated because diagrams can be "linked" or "unlinked". We will see, though, that this can be sidestepped computationally.

Take the usual Hamiltonian with some potential $U(r)$, and suppose it has eigenfunctions $\phi_k(\mathbf{r})$ and eigenvalues ϵ_k . For simplicity, suppose the Fermi system consists of only one fermion. Left alone, the system stays this way. But suppose we add a perturbation $V(\mathbf{r})$. Then $R(t)$ will consist of diagrams that show how the system interacts zero or more times with V and end up back in Fermi vacuum at the end.

Of course, with zero interactions, nothing happens. With only one interaction allowed, no particles are coming in and none can go out, so the only possibility is a fermion loop starting and ending at the interaction as we've observed in Chapter 4, roughly meaning a hole being lifted and filled instantaneously.

For a second order diagram, you could just have two first order diagrams happen back to back. But there is one other possibility: pair creation ending in pair annihilation. To see that there is no other possibility, all you need to note is that every diagram has to start with pair creation or the fermion loop, since there is no "incoming particle". The fermion loop closes itself, but when you have pair creation, they must annihilate somehow for the system to end up in vacuum. The only way to do this with one interaction left is pair annihilation.

This instance of two lower order diagrams happening to form a higher order diagram is an example of "unlinked diagrams", which is the nontriviality we discussed earlier.

Essentially, there is nothing stopping the system from performing a process that leads to vacuum, and then another one that also leads to vacuum.

You can find the third order diagrams exhaustively using similar reasoning, but I'll leave this to one of the problems. But you will see in a third order diagram that it is possible to have two particles in the same "time slice", even though our system started with one particle. The story is the same as when the diagrams violated Pauli exclusion: these parts cancel out.

A general n -th order diagram can be found by drawing n dots vertically and connecting them in such a way that every dot has at least and at most one line entering and leaving it. The drawings also have to be "topologically distinct", which is just the statement that a given diagram cannot morph into another by just moving the dots and lines around without changing the order of the dots or the directions of the lines.

With this you can find the following perturbative series for $R(t)$:

$$R = 1 + \text{diagram with 1 dot and a loop} + p \text{diagram with 2 dots and a bubble} + \dots \quad (67)$$

and you can imagine the rest, or look at a few more in Mattuck (4.20). Note that the 1 in front is to account for the case where nothing happens: you can think of it as the case where $V = 0$, so that every other diagram vanishes, in which case it is obviously true that $|R(t)|^2 = 1$.

You can then go ahead and translate these into functional form the same way we've always done, but this time there is something more interesting going on since we have the so-called "unlinked" diagrams. The goal in this next little section will be to show that the natural logarithm of the total propagator is equal to the sum of all the linked diagrams.

To show this, we first consider the same diagram repeated n times, like so:

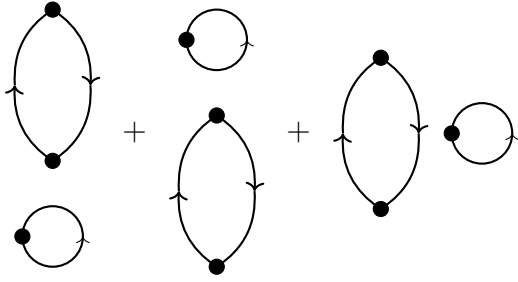
$$\begin{array}{c} t_n \text{diagram} \\ \vdots \\ t_1 \text{diagram} \end{array} \quad n \quad (68)$$

Diagram (68) ends up having the functional form:

$$V_{11}^n G_0^{-n} \int_0^t dt_1 \int_{t_1}^t dt_2 \int_{t_2}^t dt_3 \cdots \int_{t_{n-1}}^t dt_n = \frac{1}{n!} \times \text{diagram with } n \text{ dots and a loop} \quad (69)$$

where the equality can be shown by induction. Eq. (69) is actually generally true even for diagrams other than the fermion loop, and this isn't too hard to see.

When unlinked diagrams consist of parts that are not identical, then the time ordering matters: the simple fermion loop (first diagram in (67)) happening before a pair bubble (second diagram in (67)) is not the same diagram as the pair bubble happening first. Taking this as an example, we see that:



$$= \sum_p V_{p1} V_{1p} V_{11} \int \int \int dt_1 dt_2 dt_3 G_0^+(k, t_3 - t_2) G_0^-(1, t_2 - t_3) G_0^-(1, t_a - t_a) [\theta_1 + \theta_2 + \theta_3] \quad (70)$$

where I've implicitly defined t_1 as the time for the fermion loop and t_2, t_3 as the start and end times of the pair bubble. The θ_i terms are just selection rules that dictate the time ordering of each term in the sum, for example $\theta_1 = \theta_{t_3-t_2} \theta_{t_2-t_1}$. The basic takeaway is that the sum of the theta values is equal to one no matter what as long as $t_3 > t_2$, which you can reason if you stare at it.

This means all three diagrams sum up to form the entire integration range of t_1 , whose integral is independent of the integrals over t_2, t_3 . Thus, they turn into the separated form:

$$= \text{pair bubble} \times \text{fermion loop} \quad (71)$$

And again you can reason that this is generally true for all unlinked diagrams, not just for this simple case. Now we are ready to show the linked cluster theorem.

The basic idea is simple: separate all the linked diagrams into a sum, and stow them away. Then, consider the unlinked diagrams on their own. Generally, they will be combinations of the linked diagrams multiplied together, and if multiple of the same diagram appear, then in that product that diagram will be exponentiated according to how many times it appears, with the factorial out front as in (69) (or its generalization).

The idea is to group up all the unlinked diagram with n components (including multiplicity) into a single sum. This is sort of a difficult argument for higher orders so let's just consider $n = 2$, i.e. each unlinked diagram has only two components, like in (71). Then, multiply and divide by 2 on every unlinked diagram with different components, and combine them with the generalization of (69) to find:

$$R^{(2)} = \frac{1}{2!} \left[\text{pair bubble}^2 + 2 \times \text{pair bubble} \times \text{fermion loop} \right] + \dots = \frac{1}{2!} \left[\text{pair bubble} + \text{fermion loop} + \dots \right]^2 \quad (72)$$

And this will be generally true for every unlinked diagram with a general n components, i.e. it will just be the sum of all the linked diagrams raised to the n -th power, with a factor of $1/n!$ out front. It is then clear, including the 1 and linked diagrams from (67), that the total propagator is simply e^{R_l} where R_l is the linked propagator sum. From this we find the linked cluster theorem:

$$\ln R(t) = 1 + \text{[diagram: a circle with a dot on the left and a small hook on the right]} + \text{[diagram: two vertical lines connected by two arcs, with dots at the top and bottom and small hooks on the outside]} + \dots$$

$$= \sum \text{linked diagrams} \tag{73}$$

The utility of (73) is immediately clear if you look at (66): we have found a way to express the ground state energy as a sum of diagrams we can write down simply as a perturbative series, and evaluate one by one to whatever order we desire.

The goal now will be to show that this produces precisely the perturbative series seen in the usual undergraduate QM course. This is not the end goal but rather a simple check that our method works. As before, the point is that we can perform tricks like partial resummation to carry sums to infinite order.